机械与电子工程学院
College of Mechanical and Electronic Engineering

附录Ⅰ

平面图形的几何性质

Appendix Ⅰ Properties of Plane Areas

刘洪萍

	扭转（圆形、圆环形截面）	拉压
应力	$\tau_\rho = \dfrac{T\rho}{I_\rho}$ $\quad \tau_{max} = \dfrac{T_{max}}{W_t} \le [\tau]$	$\sigma = \dfrac{F_N}{A}$
变形	$\varphi = \dfrac{TL}{GI_\rho}$ $\varphi'_{max} = \dfrac{T_{max}}{GI_\rho} \le [\varphi'] \quad (\mathrm{rad/m})$ $\varphi'_{max} = \dfrac{T_{max}}{GI_\rho} \cdot \dfrac{180}{\pi} \le [\varphi'] \quad ((\circ)/\mathrm{m})$	$\Delta L = \dfrac{F_N L}{EA}$
几何量	$I_\rho = \dfrac{\pi d^4}{32}$ $\quad I_\rho = \dfrac{\pi\left(D^4 - d^4\right)}{32}$ $W_t = \dfrac{\pi d^3}{16}$ $\quad W_t = \dfrac{\pi D^3(1-\alpha^4)}{16}$	A

内容

Ⅰ-1 静矩和形心

(The first moments of the area & centroid of an area)

Ⅰ-2 惯性矩 极惯性矩 惯性积

(Moment of inertia Polar moment of inertia Product of inertia)

Ⅰ-3 平行移轴公式 (Parallel-Axis theorem)

Ⅰ-1 静矩和形心

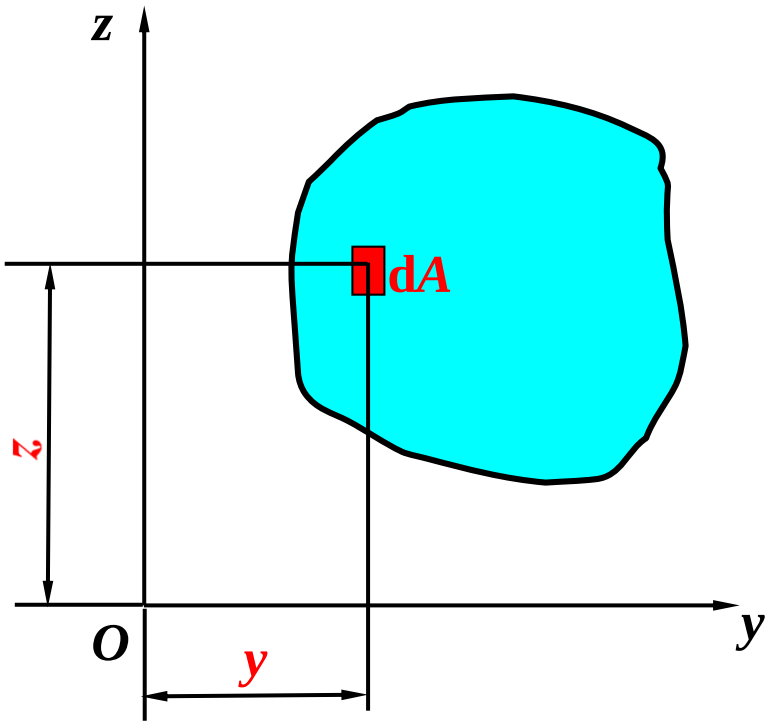
(The first moment of the area & centroid of an area)

一、静矩 (The first moment of the area）

截面对 y , z 轴的静矩为

$$\boxed{S_y = \int_A z\mathrm{d}A}$$

$$\boxed{S_z = \int_A y\mathrm{d}A}$$



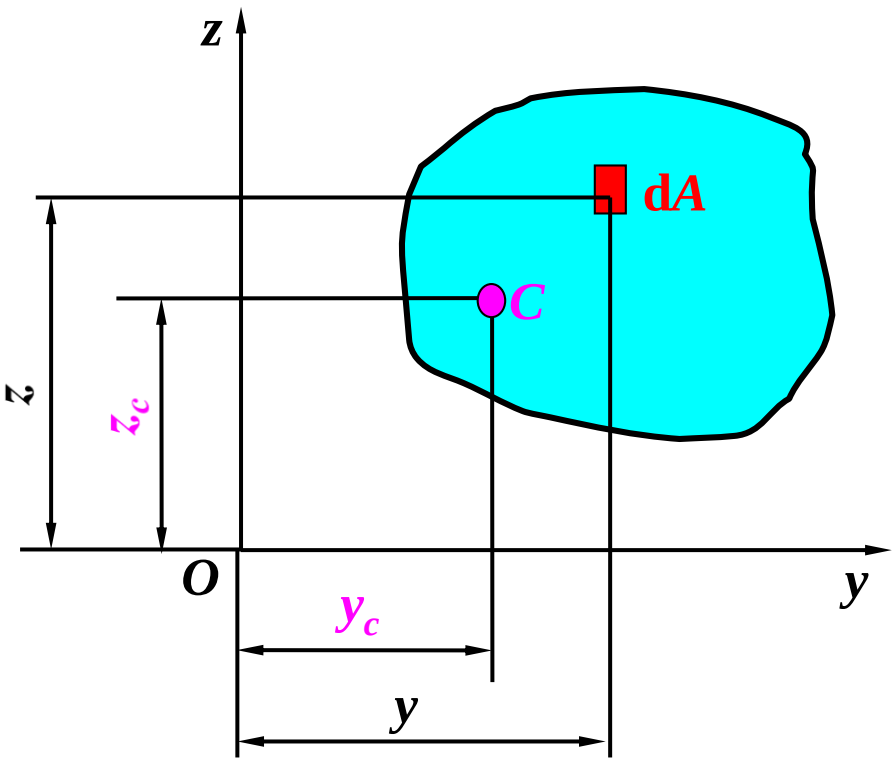
- 一次矩。
- 静矩与坐标轴有关，
  可正，可负，也可能等于零。
- 单位：m^3，mm^3。

二、形心 (Centroid of an area)

$$z_c = \frac{\int_A z dA}{A} = \frac{S_y}{A}$$

$$y_c = \frac{\int_A y dA}{A} = \frac{S_z}{A}$$

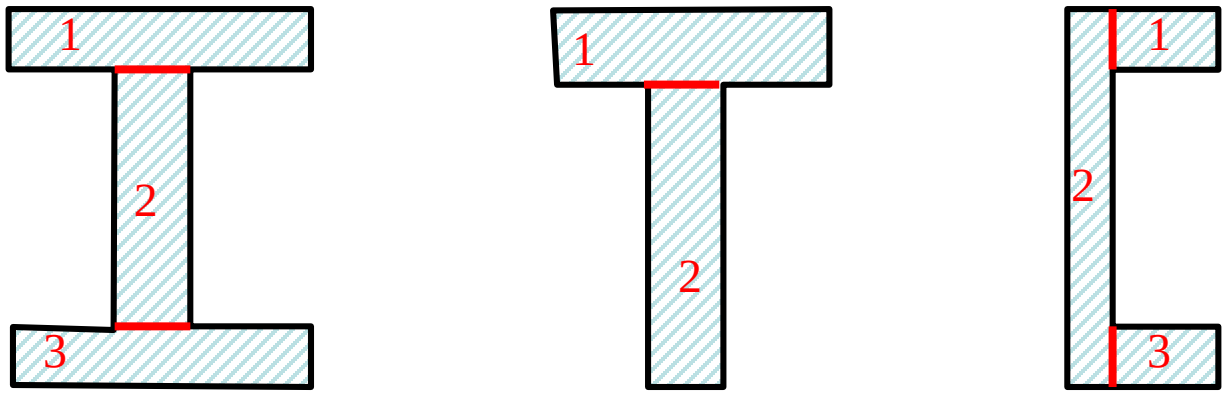
$S_y = Az_c$	$S_z = Ay_c$



- 若截面对某一轴的静矩等于零，则该轴必过形心。
- 截面对形心轴的静矩等于零。

三、组合截面的静矩和形心

(The first moments &centroid of a composite area)



组合截面静矩

$$S_y = \sum A_i z_{ci}$$

$$S_z = \sum A_i y_{ci}$$

A_i—第 i个简单截面面积

y_{ci}, z_{ci} —第 i个简单截面的形心坐标

组合截面形心

$$y_c = \frac{\sum A_i y_{ci}}{\sum A_i} \qquad z_c = \frac{\sum A_i z_{ci}}{\sum A_i}$$

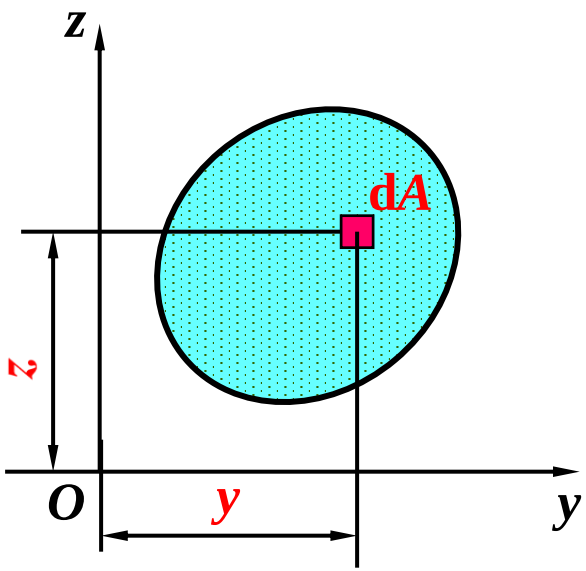
Ⅰ-2 惯性矩 极惯性矩 惯性积

(Moment of inertia　Polar moment of inertia　Product of inertia)

一、惯性矩 (Moment of inertia)

$$I_y = \int_A z^2 \mathrm{d}A$$

$$I_z = \int_A y^2 \mathrm{d}A$$



- 二次矩。
- 惯性矩与坐标轴有关，始终为正。
- 单位：m^4，mm^4。

例题1 求矩形截面对其对称轴y，z轴的惯性矩。

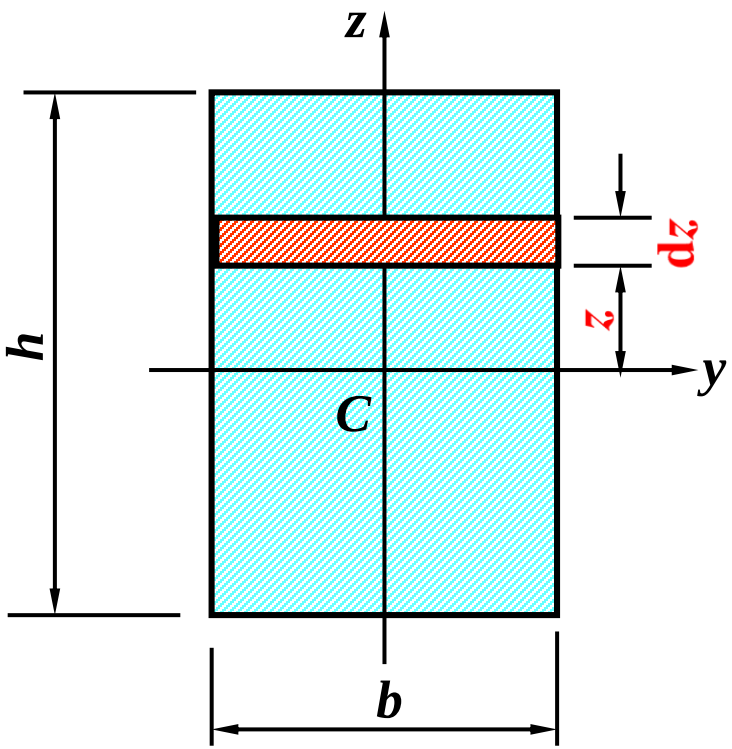
解：

$$I_y = \int_A z^2 \mathrm{d}A$$

$$\mathrm{d}A = b\mathrm{d}z$$

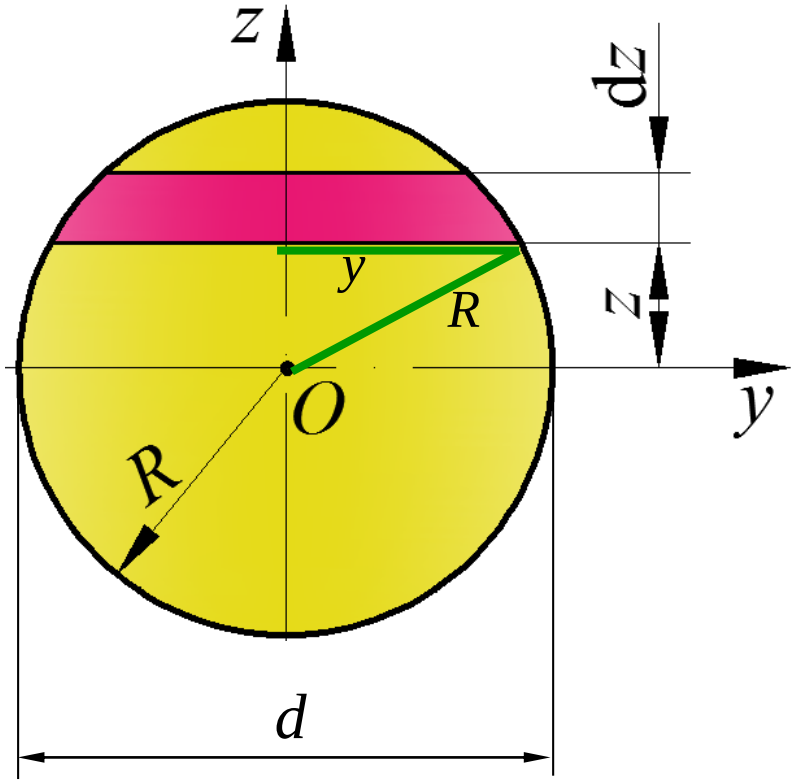
$$I_y = \int_A z^2 \mathrm{d}A = \int_{-\frac{h}{2}}^{\frac{h}{2}} bz^2 \mathrm{d}z = \boxed{\frac{bh^3}{12}}$$

$$I_z = \boxed{\frac{hb^3}{12}}$$



例题2 求圆形截面对其对称轴的惯性矩。

解：



$$\mathrm{d}A = 2y \cdot \mathrm{d}z = 2\sqrt{R^2 - z^2}\,\mathrm{d}z$$

$$I_y = \int_A z^2 \mathrm{d}A = 2\int_{-R}^{R} z^2 \cdot \sqrt{R^2 - z^2}\,\mathrm{d}z$$

$$= \frac{\pi R^4}{4} = \boxed{\frac{\pi d^4}{64}}$$

$$I_z = \boxed{\frac{\pi d^4}{64}}$$

二、惯性半径 (Radius of gyration of the area)

$$\boxed{i_y = \sqrt{\frac{I_y}{A}}} \qquad \boxed{i_z = \sqrt{\frac{I_z}{A}}}$$

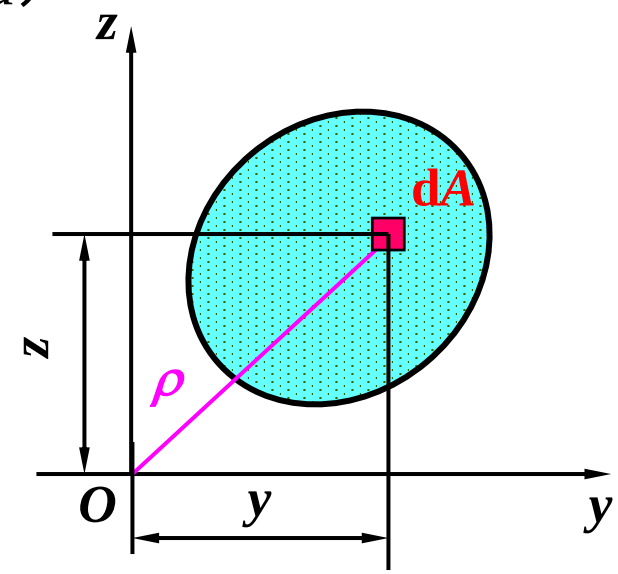
- 单位：m，mm。

如圆形截面对直径的惯性半径

$$i_y = i_z = \sqrt{\frac{\pi d^4/64}{\pi d^2/4}} = \frac{d}{4}$$

三、极惯性矩 (Polar moment of inertia）

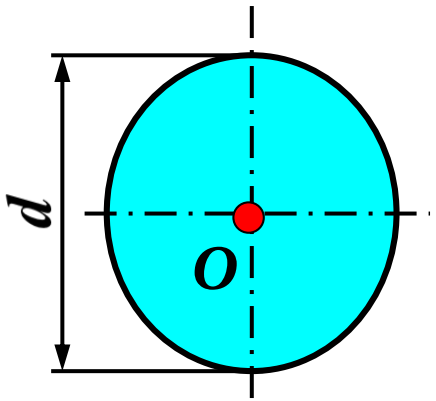
$$\boxed{I_\rho = \int_A \rho^2 \mathrm{d}A}$$



●极惯性矩与点有关，始终为正。

●单位：m^4，mm^4。

$$I_\rho = \boxed{\frac{\pi d^4}{32}}$$



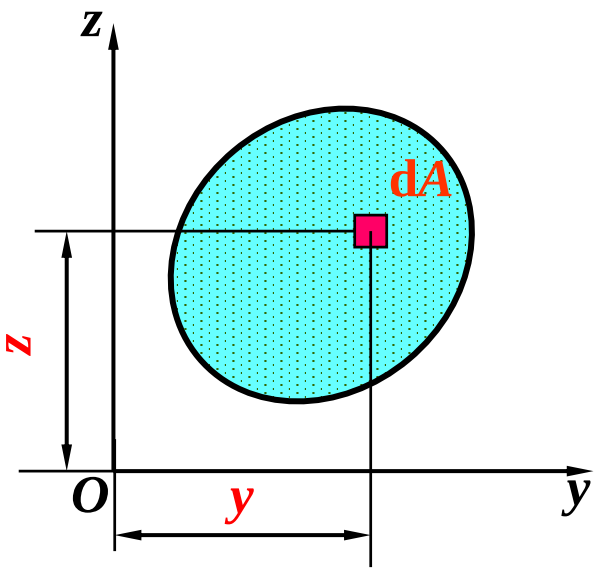
$$\rho^2 = z^2 + y^2$$

$$I_\rho = \int_A \rho^2 \mathrm{d}A = \int_A \left(y^2 + z^2\right) \mathrm{d}A = I_z + I_y$$

$$I_\rho = I_z + I_y$$

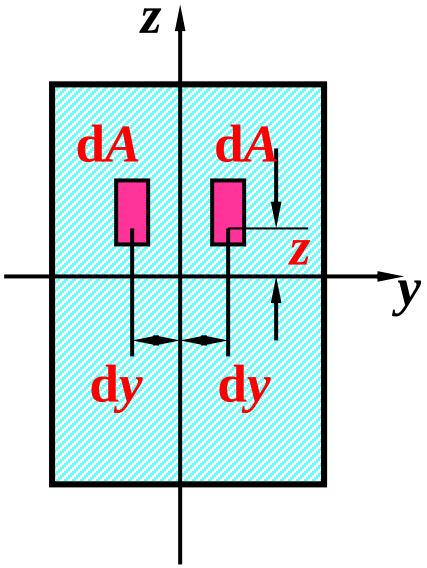
四、惯性积 (Product of inertia)

$$I_{yz} = \int_A yz\,\mathrm{d}A$$



- 惯性积与坐标轴有关，可能为正值，负值，也可能等于零；

- 若y，z 两坐标轴中有一个为截面的对称轴，则截面对y，z轴的惯性积一定等于零。



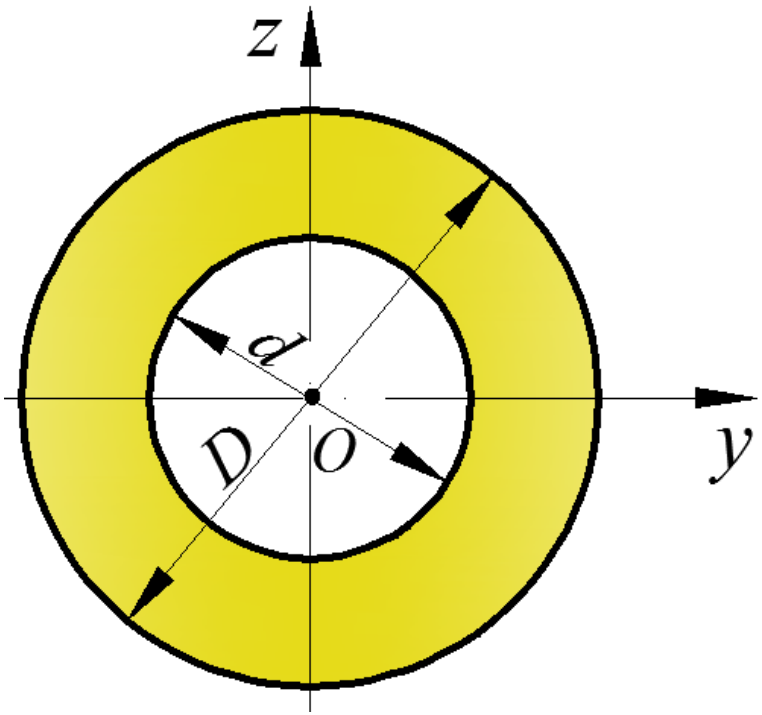
- 单位：m^4，mm^4。

五、组合截面

$$I_y = \sum I_{yi}$$

$$I_\rho = \sum I_{\rho i}$$

$$I_{yz} = \sum I_{yzi}$$



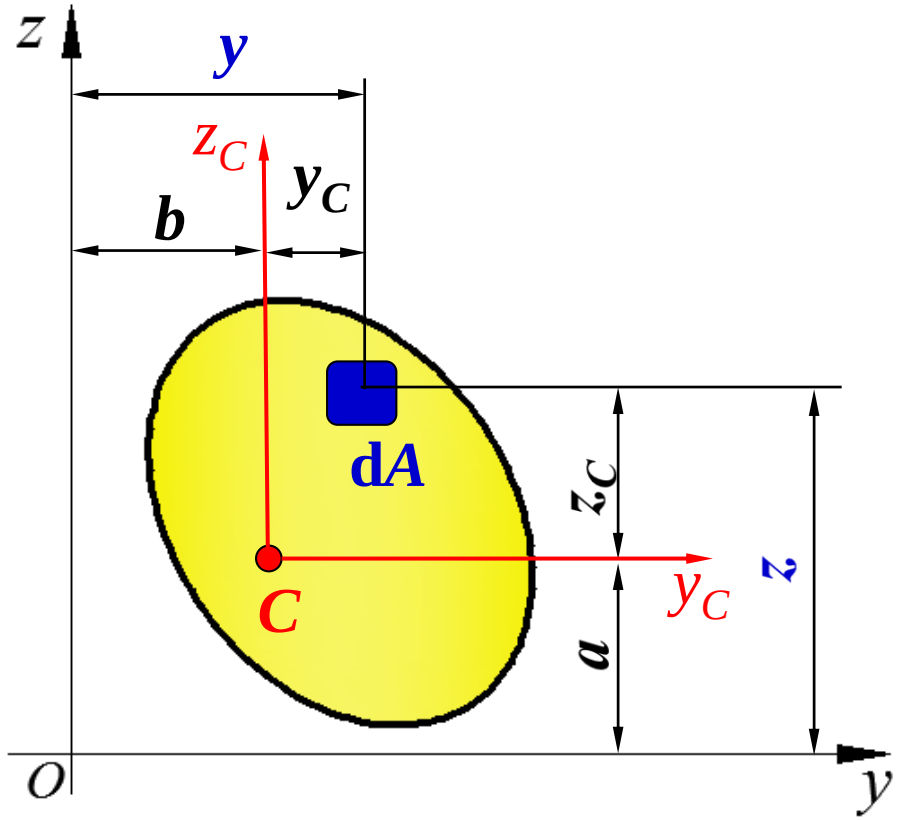
圆环形截面 $I_y = \dfrac{\pi D^4}{64} - \dfrac{\pi d^4}{64}$

πD

$$I_\rho = \frac{\pi D^4}{32} - \frac{\pi d^4}{32}$$

Ⅰ-3 平行移轴公式 (Parallel-axis theorem)

一、平行移轴公式(Parallel-Axis theorem for moment of inertia)



y、z—任意一对坐标轴

C—截面形心

a、b—形心C在oyz坐标系下的坐标

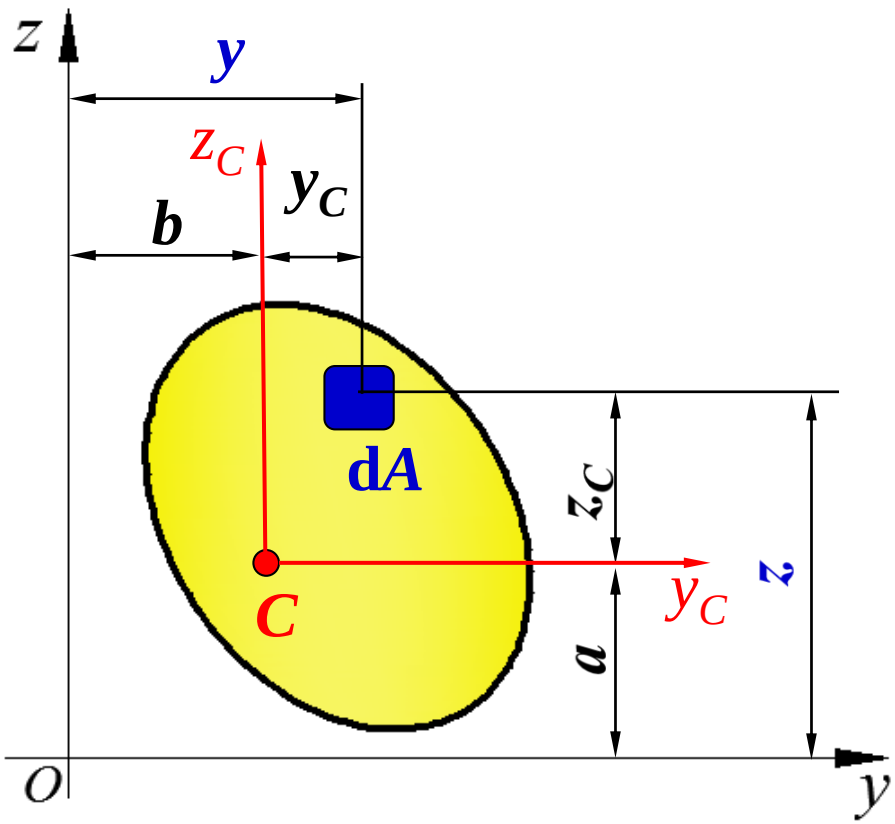
y_C, z_C—过截面的形心C且与y、z轴平行的坐标轴 (形心轴)

$$I_y = \int_A z^2 \mathrm{d}A = \int_A \left(z_C + a\right)^2 \mathrm{d}A$$

$$= \int_A \left(z_C^2 + 2az_c + a^2\right)\mathrm{d}A$$

$$= I_{y_C} + 2a\int_A z_c \mathrm{d}A + a^2 A$$

$$= I_{y_C} + a^2 A$$



$$I_z = I_{z_c} + b^2 A$$

$$I_{yz} = I_{y_c z_c} + abA$$

主惯性轴：截面对坐标轴y、z的**惯性积等于0**，则称y、z为主惯性轴。

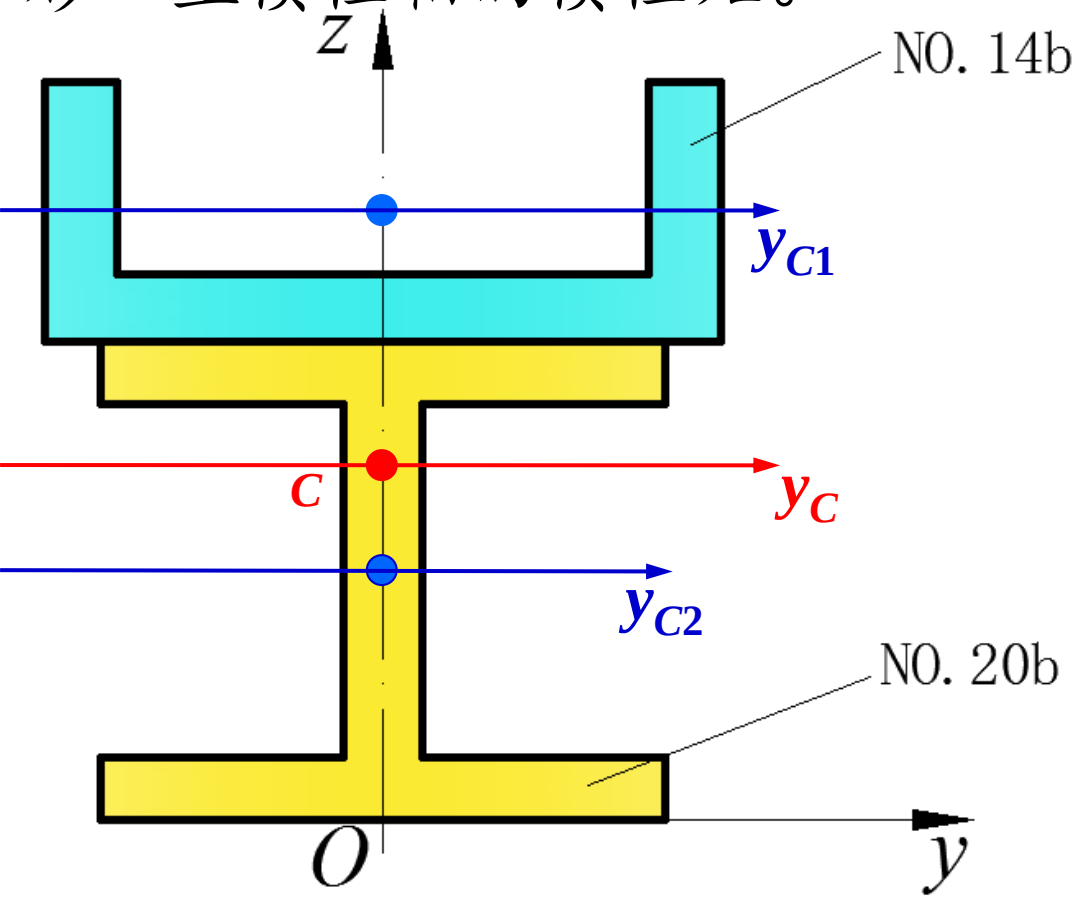
主惯性矩：截面对主惯性轴的惯性矩。

形心主惯性轴：当**一对主惯性轴的交点**与**截面的形心**重合时，则称为形心主惯性轴。

形心主惯性矩：截面对形心主惯性轴的惯性矩。

截面的对称轴一定是形心主惯性轴。

例题3 平面图形如图，求：（1）形心位置；（2）对y轴和形心轴y_c的静矩；（3）对y、z轴的惯性积。（4）对形心主惯性轴的惯性矩。



NO.14b

$$A_1 = 21.316\text{cm}^2$$
$$I_{yc1} = 61.1\text{cm}^4$$
$$I_z^1 = 609\text{cm}^4$$
$$z_{C1} = 1.67\text{cm}$$

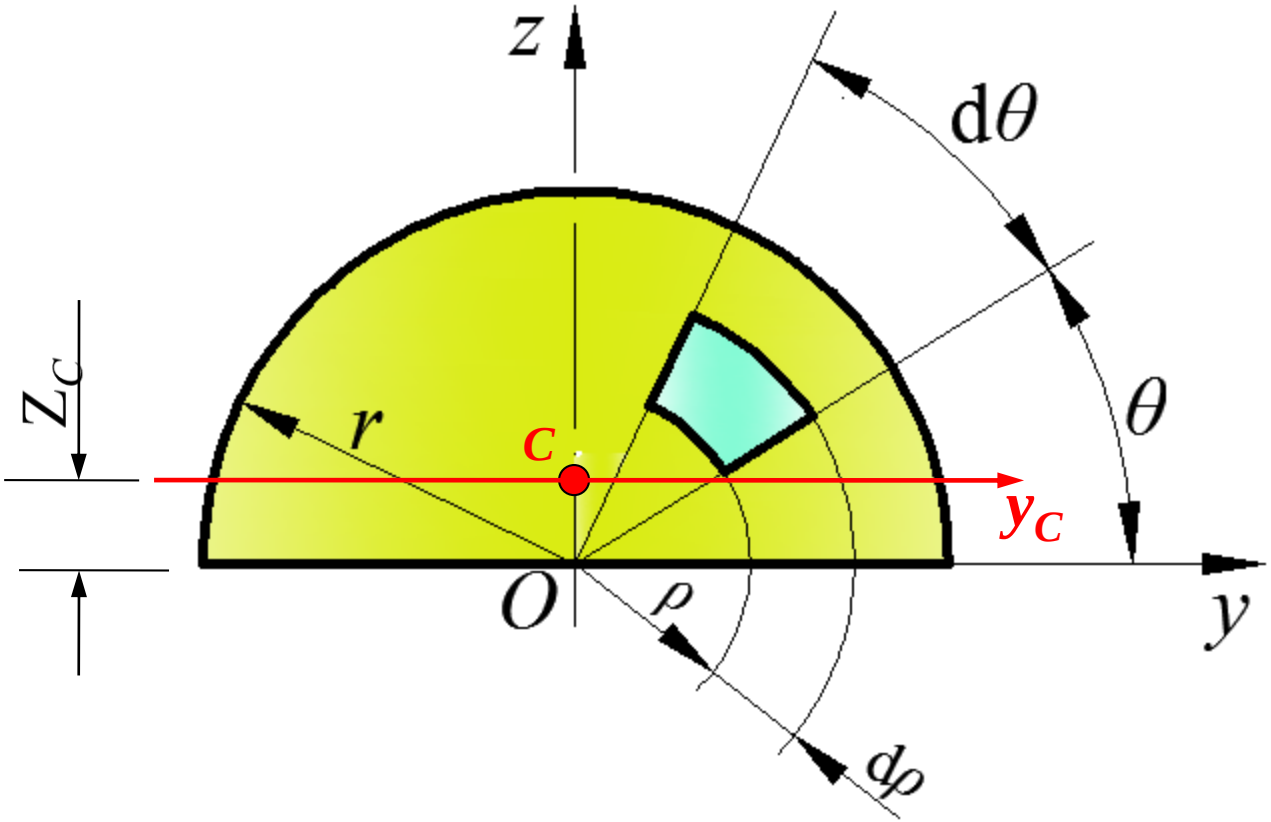
NO.20b

$$A_2 = 39.578\text{cm}^2$$
$$I_{yc2} = 2500\text{cm}^4$$
$$I_z^2 = 169\text{cm}^4$$
$$h = 20\text{cm}$$

$y_c = 0$　　$z_c = 14.1\text{cm}$　　$S_y = 857.5\text{cm}^3$　　$S_{y_C} = 0$

$I_{y_c} = 4448\text{cm}^4$　　$I_{z_c} = 778\text{cm}^4$　　$I_{yz} = 0$

例题4 平面图形如图，求：（1）形心位置；（2）对形心主惯性轴的惯性矩。



$$Z_c = \frac{4r}{3\pi}$$

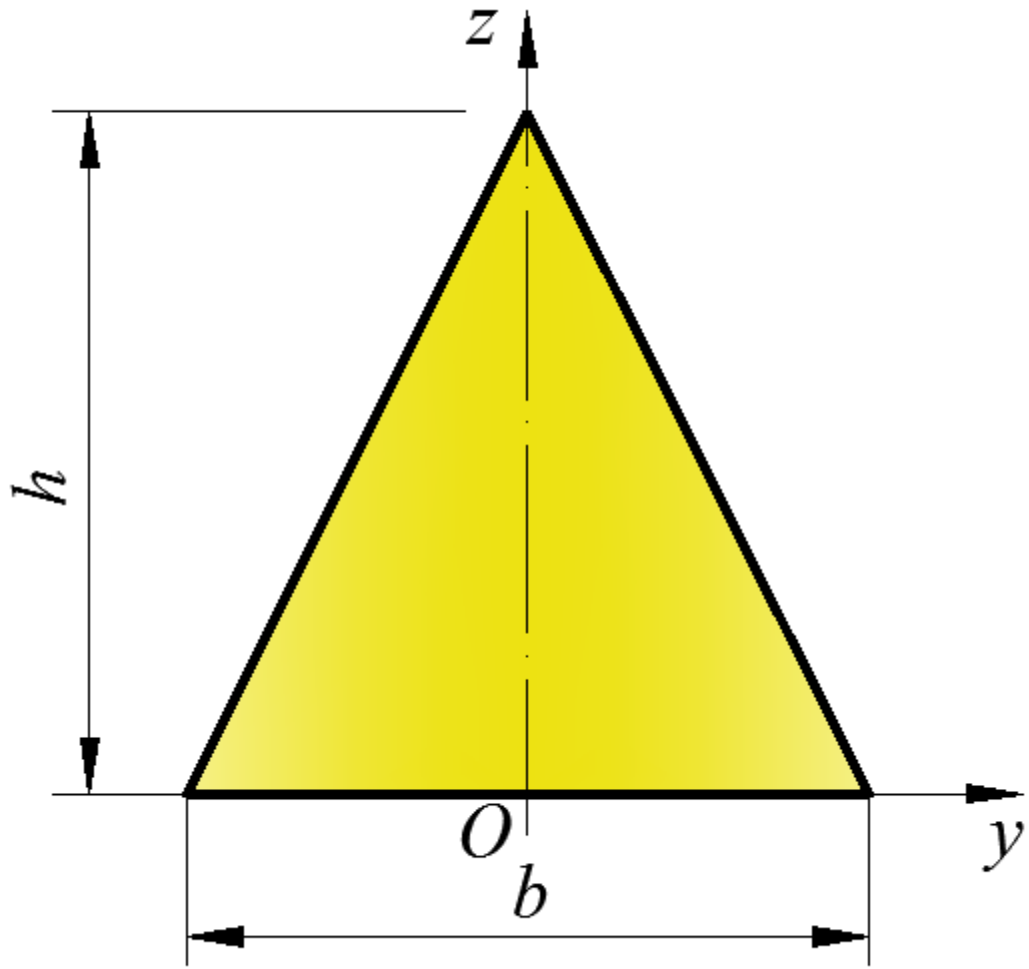
$$I_y = \frac{\pi r^4}{8}$$

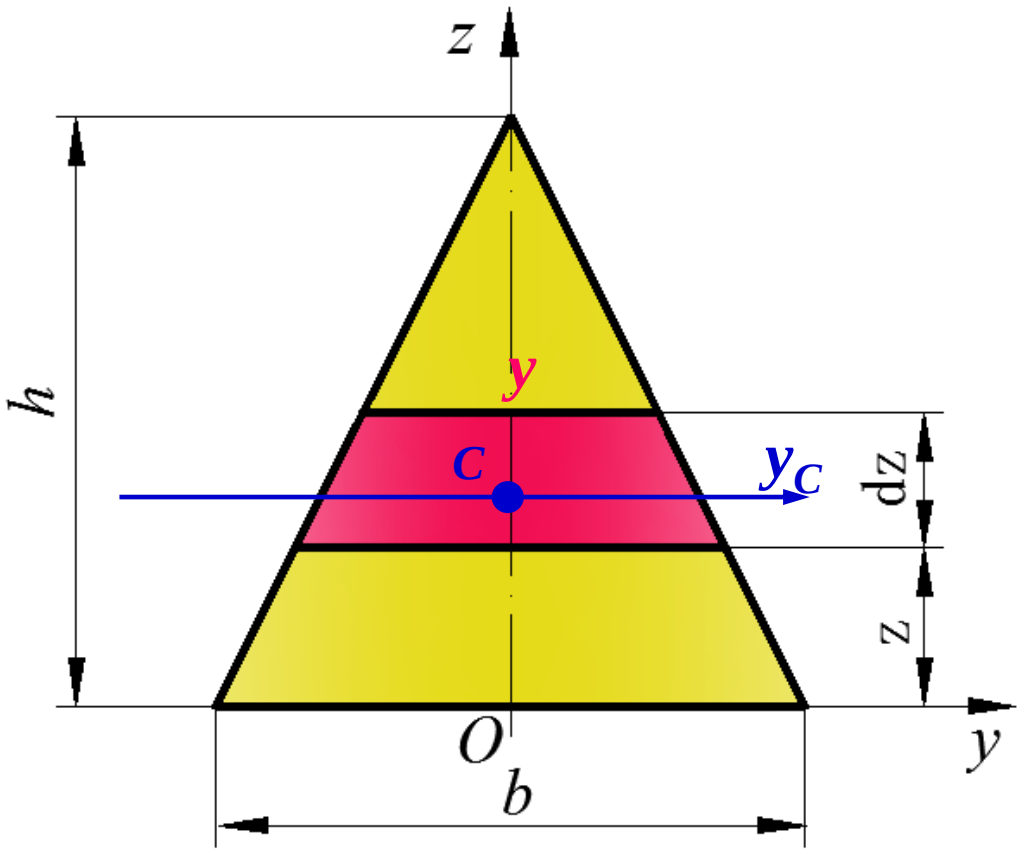
$$I_{yc} = 0.11r^4$$

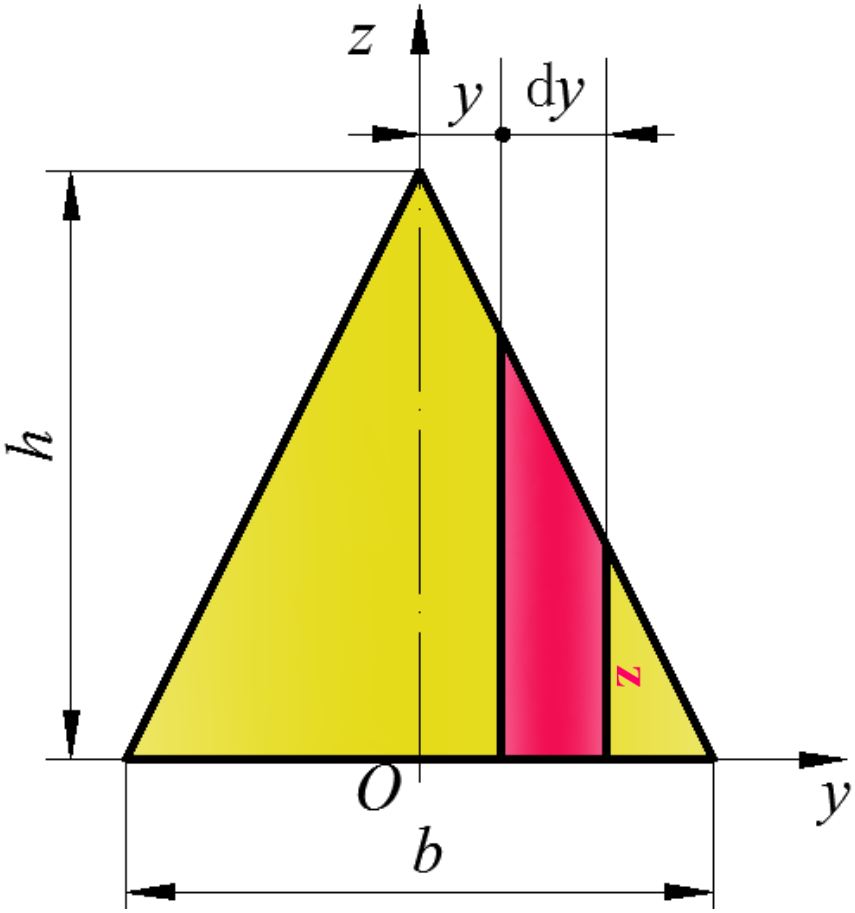
例题5 等腰三角形如图所示，底为b，高为h。

求：（1）形心位置；

（2）对形心主惯性轴的惯性矩。







$$I_y = \frac{bh^3}{12}$$

$$I_{yc} = \frac{bh^3}{36}$$

$$I_z = \frac{hb^3}{48}$$

知 识 点

静矩
形心

$$S_y = \int_A z dA = A z_c$$

惯性矩

$$I_y = \int_A z^2 dA$$

极惯性矩

$$I_\rho = \int_A \rho^2 dA$$

惯性积

$$I_{yz} = \int_A yz dA$$

惯性半径

$$i_y = \sqrt{\frac{I_y}{A}}$$

平行移轴公式

$$I_z = I_{zc} + b^2 A$$

常用值

$$I_\rho = \frac{\pi d^4}{32}$$

$$I_\rho = \frac{\pi (D^4 - d^4)}{32}$$

$$I_y = \frac{\pi d^4}{64}$$

$$I_y = \frac{\pi (D^4 - d^4)}{64}$$

$$I_y = \frac{b h^3}{12}$$

作业：

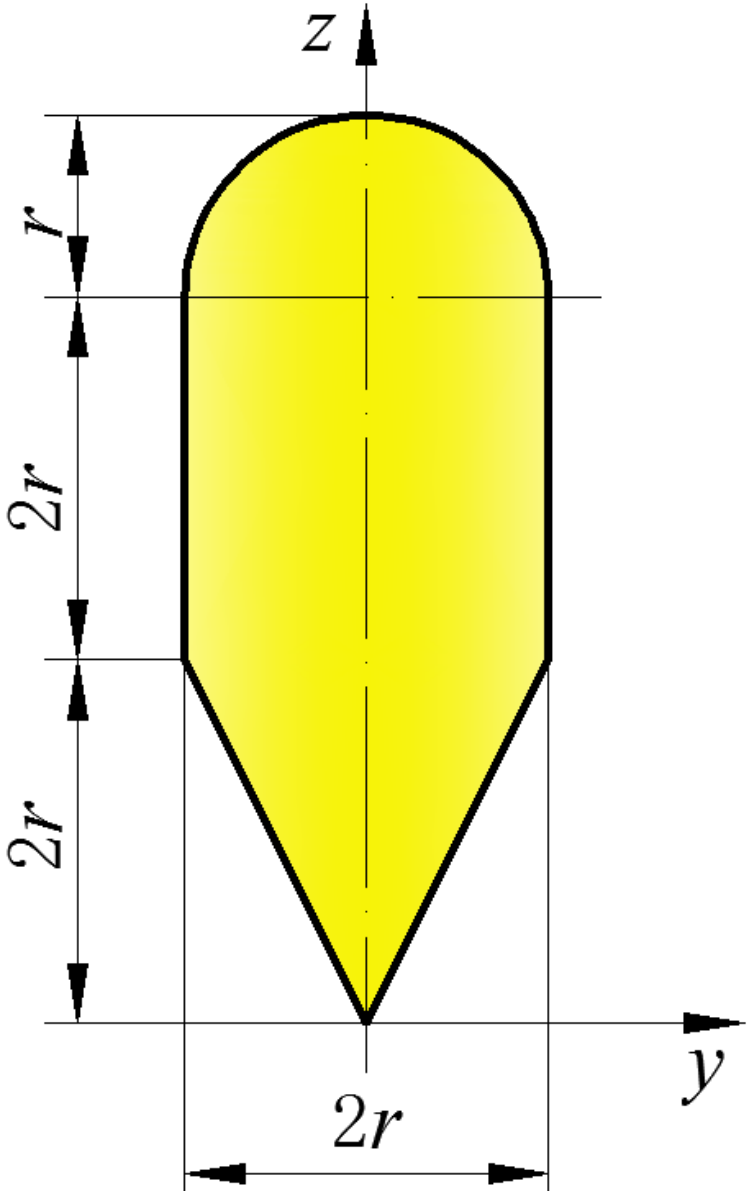
教材P351，题Ⅰ.9（a）

求：（1）形心位置；

（2）对y轴和z的静矩；

（3）对y、z轴的惯性积。

（4）对形心主惯性轴的惯性矩。



附录结束